

EA-0028 - AQELYN Engineering Archive

AQELYN - EA-0028 Engineering Archive

IS-028 - AQELYN Cloud Security Posture Management (CSPM) Intelligence Engine

****Archive ID:**** EA-0028

****Implementation Specification:**** IS-028

****Component:**** AQELYN Cloud Security Posture Management (CSPM) Intelligence Engine

****Project:**** AQELYN

****System Type:**** Cyber Security Operating Environment

****Status:**** COMPLETE

****Repository Impact:**** No top-level repository structure changes

****Breaking Changes:**** None

****Predecessor Archives:**** EA-0001 through EA-0027

****Next Specification:**** IS-029 - AQELYN SaaS Security Posture Management (SSPM) Intelligence Engine

Document Control

| Field | Value |

---|---

| Document | Engineering Archive EA-0028 |

| Specification | IS-028 - AQELYN Cloud Security Posture Management (CSPM) Intelligence Engine |

| Publication Format | Markdown, PDF, HTML, ZIP |

| Source of Truth | MD/EA-0028.md |

| Archive Rule | Implementation Specification -> Engineering Archive -> Continue |

| Repository Rule | Fixed repository structure; no redesign |

| Completion State | IS-028 complete; EA-0028 generated |

1. Engineering Context

AQELYN is being built as a modular Cyber Security Operating Environment.

Completed engineering archive chain:

| Engineering Archive | Implementation Specification | Component |

---|---|---

| EA-0001 | IS-001 | AQELYN Kernel |

| EA-0002 | IS-002 | Universal Object Model |

| EA-0003 | IS-003 | AQELYN Event Bus |

| EA-0004 | IS-004 | AQELYN Evidence Engine |

- | EA-0005 | IS-005 | AQELYN Knowledge Graph |
- | EA-0006 | IS-006 | AQELYN Trust Engine |
- | EA-0007 | IS-007 | AQELYN Mission Engine |
- | EA-0008 | IS-008 | AQELYN Workflow Engine |
- | EA-0009 | IS-009 | AQELYN Policy Engine |
- | EA-0010 | IS-010 | AQELYN Compliance & Governance Engine |
- | EA-0011 | IS-011 | AQELYN Identity & Access Governance Engine |
- | EA-0012 | IS-012 | AQELYN Asset & Configuration Governance Engine |
- | EA-0013 | IS-013 | AQELYN Risk Intelligence Engine |
- | EA-0014 | IS-014 | AQELYN Threat Intelligence Fusion Engine |
- | EA-0015 | IS-015 | AQELYN Security Operations Engine |
- | EA-0016 | IS-016 | AQELYN Digital Forensics Engine |
- | EA-0017 | IS-017 | AQELYN Threat Detection & Analytics Engine |
- | EA-0018 | IS-018 | AQELYN Automated Response & Orchestration Engine |
- | EA-0019 | IS-019 | AQELYN Security Data Lake & Telemetry Platform |
- | EA-0020 | IS-020 | AQELYN AI Decision Intelligence Engine |
- | EA-0021 | IS-021 | AQELYN Predictive Analytics & Forecasting Engine |
- | EA-0022 | IS-022 | AQELYN Executive Intelligence & Strategic Reporting Engine |
- | EA-0023 | IS-023 | AQELYN Threat Exposure & Attack Surface Management Engine |
- | EA-0024 | IS-024 | AQELYN Vulnerability Intelligence & Prioritization Engine |
- | EA-0025 | IS-025 | AQELYN Cyber Asset Discovery & Inventory Intelligence Engine |
- | EA-0026 | IS-026 | AQELYN Configuration Compliance & Drift Intelligence Engine |
- | EA-0027 | IS-027 | AQELYN Identity Threat Detection & Behavioral Analytics Engine |
- | EA-0028 | IS-028 | AQELYN Cloud Security Posture Management (CSPM) Intelligence Engine |

The fixed repository structure remains:

```
AQELYN/  
■■■■ archive/  
■■■■ blueprint/  
■■■■ docs/  
■■■■ src/  
■■■■ tests/  
■■■■ tools/  
■■■■ build/  
■■■■ releases/  
■■■■ scripts/  
■■■■ assets/  
■■■■ examples/  
■■■■ plugins/  
■■■■ sdk/  
■■■■ api/  
■■■■ README.md
```

2. IS-028 Specification Identity

Specification ID: IS-028

Name: AQELYN Cloud Security Posture Management (CSPM) Intelligence Engine

Engineering Archive Target: EA-0028

Project: AQELYN

System Type: Cyber Security Operating Environment

Status: Complete

Predecessor: IS-027 - AQELYN Identity Threat Detection & Behavioral Analytics Engine

3. Purpose

The AQELYN Cloud Security Posture Management (CSPM) Intelligence Engine provides continuous assessment of cloud environments, identifying security misconfigurations, policy violations, compliance gaps, exposed resources, excessive permissions, and cloud-native risks across multi-cloud and hybrid infrastructures.

The engine establishes continuous cloud posture intelligence by correlating cloud configuration data, identity context, workload metadata, network topology, compliance controls, and threat intelligence using explainable AI and policy-driven analytics.

It answers:

```
Which cloud resources are misconfigured?  
Which cloud accounts violate security policy?  
Which workloads are exposed?  
Which cloud identities have excessive permissions?  
Which cloud risks require immediate remediation?  
Can every cloud posture assessment be explained and audited?
```

4. Mission

The engine shall provide:

```
Continuous cloud posture assessment  
Cloud misconfiguration detection  
Cloud compliance validation  
Cloud identity posture analysis  
Cloud exposure intelligence  
Cloud risk scoring  
Mission-aware cloud reporting  
Executive cloud security summaries  
Continuous reassessment  
Cloud governance support
```

5. Scope

5.1 In Scope

```
AWS environments  
Microsoft Azure  
Google Cloud Platform  
Hybrid cloud  
Multi-cloud deployments  
Cloud workloads  
Cloud networking  
Cloud IAM  
Cloud storage  
Cloud Kubernetes services
```

5.2 Out of Scope

```
Application source code  
CI/CD pipeline execution  
Cloud billing optimization  
Cost management  
Business analytics
```

6. Dependencies

IS-028 depends on:

```
IS-001 AQELYN Kernel  
IS-002 Universal Object Model
```

IS-003 AQELYN Event Bus
 IS-004 AQELYN Evidence Engine
 IS-005 AQELYN Knowledge Graph
 IS-006 AQELYN Trust Engine
 IS-007 AQELYN Mission Engine
 IS-008 AQELYN Workflow Engine
 IS-009 AQELYN Policy Engine
 IS-010 AQELYN Compliance & Governance Engine
 IS-011 Identity & Access Governance Engine
 IS-012 Asset & Configuration Governance Engine
 IS-013 Risk Intelligence Engine
 IS-014 Threat Intelligence Fusion Engine
 IS-015 Security Operations Engine
 IS-016 Digital Forensics Engine
 IS-017 Threat Detection & Analytics Engine
 IS-018 Automated Response & Orchestration Engine
 IS-019 Security Data Lake & Telemetry Platform
 IS-020 AI Decision Intelligence Engine
 IS-021 Predictive Analytics & Forecasting Engine
 IS-022 Executive Intelligence & Strategic Reporting Engine
 IS-023 Threat Exposure & Attack Surface Management Engine
 IS-024 Vulnerability Intelligence & Prioritization Engine
 IS-025 Cyber Asset Discovery & Inventory Intelligence Engine
 IS-026 Configuration Compliance & Drift Intelligence Engine
 IS-027 Identity Threat Detection & Behavioral Analytics Engine

7. High-Level Architecture

AQELYN Cloud Security Posture Management Engine
 ■
 ■■■ Cloud Inventory Engine
 ■■■ Cloud Configuration Assessment Engine
 ■■■ Cloud Compliance Engine
 ■■■ Cloud Identity Analysis Engine
 ■■■ Cloud Exposure Intelligence Engine
 ■■■ Cloud Risk Engine
 ■■■ Knowledge Graph Connector
 ■■■ Security Data Lake Connector
 ■■■ AI Decision Connector
 ■■■ Executive Reporting Connector
 ■■■ Event Publisher

8. Functional Requirements

FR-028-001 - Cloud Inventory Assessment

The engine shall continuously discover and assess:

Cloud accounts
 Subscriptions
 Projects
 Regions
 Virtual machines
 Containers
 Storage services
 Managed databases
 Networking resources

FR-028-002 - Cloud Configuration Analysis

Continuously evaluate:

Security groups
 Firewall rules
 Encryption settings
 Logging configuration
 Public exposure
 Backup policies

Resource tagging
Configuration drift

FR-028-003 - Cloud Risk Detection

Detect:

Public storage exposure
Over-permissive IAM
Unencrypted resources
Weak networking
Shadow cloud resources
Misconfigured services

FR-028-004 - Explainable Cloud Intelligence

Every cloud assessment shall include:

Evidence references
Confidence indicators
Policy rationale
Cloud provider metadata
Historical context
Risk explanation

FR-028-005 - Governance

Support:

Cloud policy validation
Approval workflows
Version control
Auditability
Executive review

FR-028-006 - Event Publication

Publish standardized events:

cloud.resource.discovered
cloud.misconfiguration.detected
cloud.risk.updated
cloud.compliance.changed
cloud.exposure.detected
cloud.remediation.recommended

9. Non-Functional Requirements

The engine shall provide:

Continuous assessment
Enterprise scalability
Low-latency cloud analysis
Explainability
Auditability
Repository stability
Backward compatibility

10. Core Cloud Assessment Lifecycle

Cloud Resource Discovered
↓

```

Configuration Assessment
↓
Compliance Validation
↓
Exposure Analysis
↓
Risk Assessment
↓
Policy Validation
↓
Continuous Monitoring

```

11. Internal Component Architecture

The AQELYN Cloud Security Posture Management (CSPM) Intelligence Engine is implemented as a modular cloud security intelligence platform integrated with the AQELYN Kernel, Knowledge Graph, Security Data Lake, Identity Governance Engine, Configuration Compliance Engine, Risk Intelligence Engine, AI Decision Intelligence Engine, Executive Intelligence Engine, and Security Operations Engine.

```

AQELYN Cloud Security Posture Management Engine
■
■■■ Cloud Inventory Engine
■■■ Cloud Configuration Assessment Engine
■■■ Cloud Compliance Engine
■■■ Cloud Identity Analysis Engine
■■■ Cloud Exposure Intelligence Engine
■■■ Cloud Risk Engine
■■■ Knowledge Graph Connector
■■■ Security Data Lake Connector
■■■ AI Decision Connector
■■■ Executive Reporting Connector
■■■ Event Publisher

```

12. Component Specifications

12.1 Cloud Inventory Engine

Continuously discovers cloud assets.

Capabilities:

```

Cloud account discovery
Resource inventory
Subscription discovery
Project inventory
Multi-cloud normalization
Metadata collection

```

12.2 Cloud Configuration Assessment Engine

Continuously evaluates cloud configurations.

Supports:

```

Configuration assessment
Security control validation
Encryption validation
Network assessment
Logging verification
Configuration drift analysis

```

12.3 Cloud Compliance Engine

Validates cloud environments against policies.

Produces:

```
Compliance score
Control validation
Regulatory mapping
Framework correlation
Audit evidence
```

12.4 Cloud Identity Analysis Engine

Analyzes cloud IAM posture.

Produces:

```
Identity exposure
Privilege analysis
Role assessment
Permission inheritance
IAM risk scoring
```

12.5 Cloud Exposure Intelligence Engine

Analyzes attack surface.

Supports:

```
Public exposure detection
Internet-facing resources
Security group analysis
Storage exposure
Network topology intelligence
```

12.6 Cloud Risk Engine

Calculates cloud security risk.

Produces:

```
Cloud risk score
Mission impact
Business impact
Threat likelihood
Remediation priority
```

13. Universal Object Model Extensions

13.1 CloudResource

```
CloudResource:
resource_id
provider
account
region
```

13.2 CloudAssessment

```
CloudAssessment:
assessment_id
compliance_score
risk_score
generated_at
```

13.3 CloudExposure

```
CloudExposure:  
exposure_id  
exposure_type  
severity  
confidence
```

13.4 CloudRemediation

```
CloudRemediation:  
remediation_id  
recommendation  
owner  
target_date
```

14. Knowledge Graph Integration

Relationships:

```
CloudAccount  
↓  
owns  
↓  
CloudResource  
  
CloudResource  
↓  
configured_by  
↓  
Policy  
  
CloudResource  
↓  
creates  
↓  
Exposure  
  
Exposure  
↓  
contributes_to  
↓  
Risk
```

15. Event Bus Integration

15.1 Cloud Resource Events

```
cloud.resource.discovered  
cloud.resource.updated  
cloud.resource.deleted
```

15.2 Configuration Events

```
cloud.configuration.assessed  
cloud.misconfiguration.detected
```

15.3 Compliance Events

```
cloud.compliance.changed  
cloud.control.failed
```

15.4 Exposure Events


```
cloud.exposure.detected  
cloud.risk.updated  
cloud.remediation.recommended
```

16. Security Data Lake Integration

Consumes:

```
Cloud telemetry  
Configuration snapshots  
Cloud audit logs  
IAM telemetry  
Network metadata
```

17. AI Decision Intelligence Integration

Consumes:

```
Risk scores  
Exposure intelligence  
Compliance trends  
Remediation recommendations  
Confidence indicators
```

18. Risk & Threat Integration

Consumes:

```
Threat intelligence  
Mission impact  
Business criticality  
Identity risk  
Configuration compliance
```

19. Compliance Integration

Supports:

```
Cloud governance  
Regulatory mapping  
Control validation  
Audit evidence  
Executive reporting
```

20. Public APIs

20.1 Cloud Inventory API

```
GET /cloud/resources  
GET /cloud/resources/{id}
```

20.2 Cloud Assessment API

```
GET /cloud/assessments  
GET /cloud/assessments/{id}
```

20.3 Cloud Risk API

```
GET /cloud/risk
GET /cloud/risk/{id}
```

20.4 Cloud Exposure API

```
GET /cloud/exposures
POST /cloud/remediations
```

21. Repository Impact

Implementation shall use the approved repository structure.

```
AQELYN/
■■■ src/
■ ■■■ cloud_security_posture/
■■■ tests/
■ ■■■ cloud_security_posture/
■■■ docs/
■ ■■■ cloud_security_posture/
■■■ api/
■ ■■■ cloud_security_posture/
■■■ archive/
```

No top-level repository modifications are permitted.

22. Security Architecture

The AQELYN Cloud Security Posture Management (CSPM) Intelligence Engine is the trusted subsystem responsible for continuously assessing cloud environments, identifying security posture weaknesses, validating cloud compliance, detecting exposed resources, and governing cloud security intelligence.

Every cloud posture assessment shall be:

```
Explainable
Evidence-backed
Policy-governed
Cloud-aware
Risk-aware
Mission-aware
Fully auditable
Continuously reassessed
```

22.1 Security Principles

```
Zero Trust
Defense in Depth
Least Privilege
Continuous Cloud Assessment
Secure by Design
Policy Enforcement
Explainable Cloud Intelligence
Continuous Monitoring
```

22.2 Authorization Model

Supported operational roles:

```
Cloud Security Administrator
Cloud Architect
Security Administrator
```

```
SOC Analyst
Compliance Officer
Mission Owner
Cloud Operations Engineer
Executive Reviewer
```

All cloud posture assessments and remediation decisions shall be governed through the AQELYN Policy Engine.

22.3 Cloud Assessment Integrity

Cloud assessments shall maintain:

```
Unique assessment identifier
Cloud resource reference
Cloud provider metadata
Policy references
Control mappings
Risk score
Confidence indicators
Remediation guidance
Audit trail
```

Assessment history shall be append-only.

22.4 Cloud Resource Integrity

Cloud resources shall maintain:

```
Resource identifier
Provider
Account or subscription
Region
Ownership metadata
Configuration metadata
Exposure state
Validation timestamp
Audit record
```

No cloud resource shall be considered authoritative without provider metadata, source lineage, and validation state.

23. Cloud Security Lifecycle

23.1 Assessment Lifecycle

```
Cloud Resource Discovered
↓
Configuration Assessment
↓
Compliance Validation
↓
Exposure Analysis
↓
Risk Assessment
↓
Continuous Monitoring
```

23.2 Remediation Lifecycle

```
Issue Detected
↓
Risk Prioritized
↓
Remediation Generated
↓
Approval Granted
↓
```

Remediation Applied
↓
Validation Completed

23.3 Audit Lifecycle

Assessment Created
↓
Evidence Linked
↓
Policy Validated
↓
Audit Stored

24. Continuous Cloud Operations

The engine continuously evaluates:

Cloud resources
Cloud identities
Cloud networking
Cloud storage
Container services
Serverless workloads
Cloud compliance
Cloud exposure

25. Performance Requirements

The engine shall support:

Continuous assessment
Low-latency cloud analysis
Enterprise-scale cloud environments
Concurrent cloud assessments
High availability
Continuous operation

26. Scalability Requirements

The engine shall scale to support:

Global enterprises
Hybrid cloud
Multi-cloud providers
Millions of cloud resources
Distributed cloud assessments
Long-term posture history

27. Audit Requirements

Every cloud posture operation shall generate immutable audit records.

Audit events include:

Cloud resource discovered
Cloud assessment completed
Cloud misconfiguration detected

Compliance status changed
Exposure detected
Remediation completed

28. Failure Handling

28.1 Assessment Failure

Assessment retried
Failure recorded
Administrator notified

28.2 Compliance Failure

Validation retried
Previous assessment retained
Audit generated

28.3 Exposure Analysis Failure

Analysis repeated
Manual review initiated
Audit recorded

28.4 Policy Failure

Cloud approval blocked
Policy violation recorded
Manual approval required

29. Testing Strategy

29.1 Unit Testing

Validate:

Cloud Inventory Engine
Cloud Configuration Assessment Engine
Cloud Compliance Engine
Cloud Identity Analysis Engine
Cloud Exposure Intelligence Engine
Cloud Risk Engine

29.2 Integration Testing

Verify interaction with:

Kernel
Knowledge Graph
Security Data Lake
Identity Governance Engine
Configuration Compliance Engine
Risk Intelligence Engine
AI Decision Engine
Executive Reporting Engine
Security Operations Engine

29.3 System Testing

Validate:

- Cloud inventory
- Configuration assessment
- Compliance validation
- Exposure analysis
- Risk scoring
- Audit generation

29.4 Security Testing

Verify:

- Authorization
- Policy enforcement
- Explainability
- Audit logging
- Cloud assessment integrity

29.5 Regression Testing

Verify IS-001 through IS-027 remain unaffected.

30. Acceptance Criteria

IS-028 is complete when:

- Cloud Inventory Engine implemented
- Cloud Configuration Assessment Engine implemented
- Cloud Compliance Engine implemented
- Cloud Identity Analysis Engine implemented
- Cloud Exposure Intelligence Engine implemented
- Cloud Risk Engine implemented
- Repository unchanged
- Testing documented

31. Repository Validation

Repository structure remains unchanged.

```
AQELYN/  
■■■ src/cloud_security_posture/  
■■■ tests/cloud_security_posture/  
■■■ docs/cloud_security_posture/  
■■■ api/cloud_security_posture/  
■■■ archive/
```

No top-level repository modifications are permitted.

32. Engineering Summary

IS-028 introduces the AQELYN Cloud Security Posture Management (CSPM) Intelligence Engine, providing enterprise-scale cloud security posture assessment, cloud compliance validation, cloud identity analysis, exposure intelligence, explainable cloud risk scoring, and continuous cloud governance.

Major capabilities include:

- Cloud Resource Discovery
- Cloud Configuration Assessment
- Cloud Compliance Validation
- Cloud Identity Analysis

Cloud Exposure Intelligence
Cloud Risk Scoring
Mission-Aware Cloud Reporting
Executive Cloud Security Reporting
Continuous Cloud Monitoring
Cloud Governance

The engine integrates with all previously completed AQELYN engines while preserving repository stability, modularity, and backward compatibility.

33. Specification Status

Specification ID : IS-028
Title : AQELYN Cloud Security Posture Management (CSPM) Intelligence Engine
Status : COMPLETE
Engineering Archive : READY FOR GENERATION
Next Artifact : EA-0028

34. EA-0028 Engineering Objective

The objective of IS-028 was to introduce a dedicated Cloud Security Posture Management Intelligence Engine that enables AQELYN to continuously assess cloud resources, detect misconfigurations, validate cloud compliance, analyze cloud identity posture, detect exposed resources, score cloud risk, and support cloud remediation.

The engine extends AQELYN from identity threat detection into continuous cloud security posture governance.

35. EA-0028 Engineering Summary

The implementation specification defines a modular subsystem responsible for:

Cloud inventory discovery
Cloud configuration assessment
Cloud compliance validation
Cloud identity posture analysis
Cloud exposure intelligence
Cloud risk scoring
Cloud remediation recommendations
Knowledge Graph integration
Security Data Lake integration
Risk and threat intelligence integration
AI Decision integration
Executive reporting integration
Event publishing

36. Major Engineering Decisions

36.1 Decision 1 - Dedicated CSPM Intelligence Engine

Cloud posture assessment is implemented as a standalone engine rather than embedded in asset inventory, configuration compliance, or security operations.

Rationale:

Clear separation of cloud posture from general configuration compliance.
Independent lifecycle and governance.
Better support for multi-cloud normalization, cloud IAM analysis, and cloud exposure intelligence.

Improved traceability for cloud assessment decisions.

36.2 Decision 2 - Multi-Cloud Normalization Is Required

Cloud resources are normalized across AWS, Microsoft Azure, Google Cloud Platform, hybrid cloud, and multi-cloud deployments.

Benefits:

Executive reporting can compare cloud posture across providers.
Cloud risk can be scored consistently.
Knowledge Graph relationships remain portable.

36.3 Decision 3 - Cloud Identity Analysis Is First-Class

Cloud identity posture analysis is modeled directly because excessive permissions and inheritance are a major cloud risk factor.

Benefits:

Over-permissive IAM can be detected.
Cloud identity exposure becomes visible.
Cloud risk scoring includes identity context.

36.4 Decision 4 - Cloud Assessments Are Evidence-Backed

Every cloud posture assessment includes evidence references, confidence indicators, policy rationale, cloud provider metadata, historical context, and risk explanation.

Benefits:

Cloud findings become defensible.
Security teams can explain remediation decisions.
Compliance teams can audit posture changes.

36.5 Decision 5 - Event-Driven Cloud Lifecycle

Cloud resource, configuration, compliance, exposure, risk, and remediation events are published through the AQELYN Event Bus.

Examples include:

```
cloud.resource.discovered
cloud.resource.updated
cloud.resource.deleted
cloud.configuration.assessed
cloud.misconfiguration.detected
cloud.compliance.changed
cloud.control.failed
cloud.exposure.detected
cloud.risk.updated
cloud.remediation.recommended
```

36.6 Decision 6 - Universal Object Model Extension

New domain objects introduced include:

```
CloudResource
CloudAssessment
CloudExposure
CloudRemediation
```


37. Architectural Integration Summary

| Engine | Integration |

|---|---|

| IS-001 Kernel | Runtime lifecycle and service registration |

| IS-002 Universal Object Model | Cloud resource, assessment, exposure, remediation objects |

| IS-003 Event Bus | Cloud resource, configuration, compliance, exposure events |

| IS-004 Evidence Engine | Evidence references and cloud assessment support |

| IS-005 Knowledge Graph | Cloud account, resource, policy, exposure, risk relationships |

| IS-006 Trust Engine | Data confidence and cloud assessment trust |

| IS-007 Mission Engine | Mission-aware cloud risk prioritization |

| IS-008 Workflow Engine | Remediation and approval workflows |

| IS-009 Policy Engine | Cloud policy validation and governance |

| IS-010 Compliance Engine | Regulatory mapping and cloud audit evidence |

| IS-011 Identity Governance Engine | Cloud identity and IAM context |

| IS-013 Risk Intelligence Engine | Cloud risk scoring and business impact |

| IS-014 Threat Intelligence Engine | Threat context for cloud exposures |

| IS-015 Security Operations Engine | Investigation and response workflow |

| IS-019 Security Data Lake | Cloud telemetry, audit logs, configuration snapshots |

| IS-020 AI Decision Engine | Remediation recommendations and learning insights |

| IS-022 Executive Reporting Engine | Executive cloud security summaries |

| IS-026 Configuration Compliance Engine | Configuration posture and drift context |

| IS-027 Identity Threat Engine | Identity risk and behavior context |

No existing engine required redesign.

38. Repository Impact Summary

Repository structure remains unchanged.

Implementation is expected within existing project directories, including:

```
AQELYN/
■■■ src/cloud_security_posture/
■■■ tests/cloud_security_posture/
■■■ api/cloud_security_posture/
■■■ docs/cloud_security_posture/
■■■ archive/
```

No top-level directories were added, removed, or renamed.

39. Security Impact Summary

The specification introduces cloud-posture-specific security controls:

```
Policy-governed cloud assessments
Evidence-backed cloud findings
Cloud provider metadata lineage
Continuous cloud monitoring
Cloud IAM posture analysis
Cloud exposure intelligence
Cloud compliance validation
```

Cloud remediation traceability
Role-authorized cloud posture administration

No reduction in the security posture of existing components was identified.

40. Capabilities Added

The engine enables AQELYN to support:

Continuous cloud posture assessment
Cloud misconfiguration detection
Cloud compliance validation
Cloud identity posture analysis
Cloud exposure intelligence
Cloud risk scoring
Mission-aware cloud reporting
Executive cloud security summaries
Continuous reassessment
Cloud governance support

41. Risks Identified

| Risk | Mitigation |

|---|---|

| Cloud provider blind spots | Multi-cloud normalization and telemetry ingestion |

| Excessive cloud permissions | Cloud Identity Analysis Engine |

| Public cloud exposure | Cloud Exposure Intelligence Engine |

| Misconfiguration drift | Configuration assessment and drift integration |

| Cloud compliance gaps | Cloud Compliance Engine and control mapping |

| Alert overload | Cloud risk scoring and prioritization |

| Unauthorized remediation | Policy enforcement and approval workflows |

| Weak auditability | Evidence-backed assessments and immutable audit trail |

No critical architectural risks were identified that require redesign.

42. Verification Summary

The specification defines verification for:

Unit testing
Integration testing
System testing
Security testing
Regression testing

Acceptance criteria cover cloud inventory, configuration assessment, compliance, identity analysis, exposure intelligence, risk engine, repository validation, and testing documentation.

43. Engineering Principles Confirmed

The implementation complies with established AQELYN principles:

Modular architecture
 Event-driven communication
 Immutable evidence references
 Traceability
 Explainability
 Security by design
 Repository stability
 Backward compatibility
 Governance-before-archive discipline

44. Dependencies

Required:

EA-0001 through EA-0027
 IS-001 through IS-028

Enables:

IS-029 and subsequent SaaS posture, supply chain intelligence, and cyber resilience components

45. Completion Record

Engineering Archive : EA-0028
 Implementation Specification : IS-028
 Title : AQELYN Cloud Security Posture Management (CSPM) Intelligence Engine
 Engineering Status : COMPLETE
 Repository Status : UNCHANGED
 Architecture Status : EXTENDED
 Backward Compatibility : MAINTAINED

46. Archive Index Update

EA-0001 IS-001 AQELYN Kernel
 EA-0002 IS-002 Universal Object Model
 EA-0003 IS-003 AQELYN Event Bus
 EA-0004 IS-004 AQELYN Evidence Engine
 EA-0005 IS-005 AQELYN Knowledge Graph
 EA-0006 IS-006 AQELYN Trust Engine
 EA-0007 IS-007 AQELYN Mission Engine
 EA-0008 IS-008 AQELYN Workflow Engine
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 EA-0012 IS-012 AQELYN Asset & Configuration Governance Engine
 EA-0013 IS-013 AQELYN Risk Intelligence Engine
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 EA-0018 IS-018 AQELYN Automated Response & Orchestration Engine
 EA-0019 IS-019 AQELYN Security Data Lake & Telemetry Platform
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 EA-0025 IS-025 AQELYN Cyber Asset Discovery & Inventory Intelligence Engine
 EA-0026 IS-026 AQELYN Configuration Compliance & Drift Intelligence Engine
 EA-0027 IS-027 AQELYN Identity Threat Detection & Behavioral Analytics Engine
 EA-0028 IS-028 AQELYN Cloud Security Posture Management (CSPM) Intelligence Engine

47. Engineering Phase Status

Completed Engineering Archives : EA-0001 through EA-0028

Current Status:
EA-0028 COMPLETE

Next Implementation Specification:
IS-029 - AQELYN SaaS Security Posture Management (SSPM) Intelligence Engine

48. Engineering Archive Publication Standard

EA-0028 follows the AQELYN Engineering Archive Publication Standard.

Required package structure:

```
EA-xxxx/  
■  
■■■■ diagrams/  
■ ■■■■ Architecture.svg  
■ ■■■■ Component.svg  
■ ■■■■ Workflow.svg  
■ ■■■■ EventFlow.svg  
■ ■■■■ Integration.svg  
■  
■■■■ examples/  
■ ■■■■ example_*.md  
■  
■■■■ HTML/  
■ ■■■■ index.html  
■ ■■■■ styles.css  
■ ■■■■ assets/  
■  
■■■■ images/  
■  
■■■■ journal/  
■ ■■■■ Engineering_Journal.md  
■  
■■■■ manifest/  
■ ■■■■ manifest.json  
■  
■■■■ MD/  
■ ■■■■ EA-xxxx.md  
■  
■■■■ PDF/  
■ ■■■■ EA-xxxx.pdf  
■  
■■■■ requirements/  
■ ■■■■ Requirements_Matrix.md  
■  
■■■■ traceability/  
■ ■■■■ Traceability_Matrix.md  
■  
■■■■ README.md
```

The master Markdown is the source of truth. The PDF and HTML are generated from the same master Markdown and must not omit sections.

49. Requirements Matrix

Requirement ID Requirement Evidence in Archive Status				
--- --- --- ---				
FR-028-001	Continuously discover and assess cloud inventory	Sections 8, 12, 23	Complete	
FR-028-002	Analyze cloud configurations	Sections 8, 12, 23	Complete	
FR-028-003	Detect cloud risks	Sections 8, 12, 23	Complete	
FR-028-004	Provide explainable cloud intelligence	Sections 8, 22, 36	Complete	
FR-028-005	Support governance	Sections 8, 22, 23	Complete	

| FR-028-006 | Publish cloud events | Sections 8, 15, 36 | Complete |

| NFR-028-001 | Continuous assessment | Sections 9, 24 | Complete |

| NFR-028-002 | Enterprise scalability | Sections 9, 25, 26 | Complete |

| NFR-028-003 | Low-latency cloud analysis | Sections 9, 25 | Complete |

| NFR-028-004 | Explainability | Sections 9, 22, 36 | Complete |

| NFR-028-005 | Auditability | Sections 9, 27, 39 | Complete |

| NFR-028-006 | Repository stability | Sections 21, 31, 38 | Complete |

50. Traceability Matrix

| Source | Target | Relationship |

|---|---|---|

| IS-028 Purpose | EA-0028 Objective | Defines why the engine exists |

| Cloud Inventory Engine | FR-028-001 | Discovers cloud resources |

| Cloud Configuration Assessment Engine | FR-028-002 | Evaluates cloud configuration |

| Cloud Compliance Engine | Compliance validation | Maps controls and regulations |

| Cloud Identity Analysis Engine | Cloud IAM posture | Analyzes cloud identity and privilege |

| Cloud Exposure Intelligence Engine | FR-028-003 | Detects cloud exposure |

| Cloud Risk Engine | Risk assessment | Calculates cloud risk and priority |

| Event Publisher | FR-028-006 | Publishes cloud posture events |

| Security Data Lake Integration | Cloud telemetry | Supplies logs and snapshots |

| AI Decision Integration | Remediation recommendations | Supplies confidence and recommendations |

| Risk & Threat Integration | Cloud threat context | Supplies risk and mission impact |

| Compliance Integration | Cloud governance | Supports audit and reporting |

| Repository Validation | Repository Standard | Confirms no top-level redesign |

51. Engineering Journal

Journal Entry - EA-0028

EA-0028 was created to archive completion of IS-028 - AQELYN Cloud Security Posture Management (CSPM) Intelligence Engine.

The archive records the expansion of AQELYN into cloud security posture management. IS-028 defines the structure needed to discover cloud resources, assess cloud configurations, validate cloud compliance, analyze cloud identity posture, detect cloud exposure, calculate cloud risk, and publish cloud posture lifecycle events.

The engineering design preserves the fixed AQELYN repository structure and maintains backward compatibility with previously completed engines.

Lessons Learned

Cloud security posture must be modeled separately from generic configuration compliance because cloud providers introduce provider-specific identity, networking, storage, resource, and compliance semantics that require cloud-aware analytics.

Governance Note

EA-0028 follows the master-document publication workflow. The Markdown file is the authoritative source, and PDF/HTML representations are generated from the same content.

52. Examples

52.1 Example Cloud Resource

```
resource_id: CLOUD-RES-0001
provider: aws
account: production-security
region: eu-west-1
resource_type: object_storage
exposure_state: public
```

52.2 Example Cloud Assessment

```
assessment_id: CLOUD-ASSESS-1001
compliance_score: 72
risk_score: 88
generated_at: 2026-07-07T12:00:00Z
```

52.3 Example Cloud Exposure

```
exposure_id: CLOUD-EXP-2001
exposure_type: public_storage
severity: high
confidence: 0.91
```

52.4 Example Cloud Event

```
{
  "event_type": "cloud.misconfiguration.detected",
  "resource_id": "CLOUD-RES-0001",
  "risk_score": 88,
  "source_engine": "aqelyn_cloud_security_posture_management_engine"
}
```

53. Manifest Summary

Archive contents include:

```
README.md
MD/EA-0028.md
PDF/EA-0028.pdf
HTML/index.html
HTML/styles.css
manifest/manifest.json
requirements/Requirements_Matrix.md
traceability/Traceability_Matrix.md
journal/Engineering_Journal.md
diagrams/Architecture.svg
diagrams/Component.svg
diagrams/Workflow.svg
diagrams/EventFlow.svg
diagrams/Integration.svg
examples/example_cloud_security_posture.md
```

54. Final Archive Statement

EA-0028 is the Engineering Archive for IS-028 - AQELYN Cloud Security Posture Management (CSPM) Intelligence Engine.

It records the completed specification, the architectural decisions, the integration model, the repository impact, the risk posture, verification requirements, acceptance criteria, archive index update, and the engineering publication standard.

EA-0028 COMPLETE

IS-028 COMPLETE

NEXT: IS-029